

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
PULCHOWK CAMPUS

A PROJECT PROPOSAL ON

NLP-Based Sentiment and Topic Analysis of Customer Reviews

SIC AI CAPSTONE PROJECT

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1 Introduction

E-commerce platforms in Nepal, such as Daraz, generate a large and continuously growing volume of customer reviews on a daily basis. These reviews contain rich, unstructured feedback about product quality, pricing, delivery experience, and seller service. However, this feedback is written informally, often mixing English with romanized Nepali (Nepali written using Roman/English script), and typically expresses opinions about multiple aspects of a product within a single review.

Manually reading through thousands of such reviews to understand customer sentiment and recurring concerns is impractical for sellers and platform administrators. Natural Language Processing (NLP) offers automated techniques to extract sentiment polarity and dominant discussion topics from raw review text, converting unstructured customer feedback into structured, actionable insight.

This project proposes to build an NLP-based system that performs sentiment analysis and topic modeling on a dataset of customer reviews scraped from Daraz Nepal. Because the dataset is expected to contain a code-mixed combination of English and romanized Nepali text, the project places particular emphasis on handling this linguistic characteristic, which is common in South Asian e-commerce platforms but is underexplored compared to purely English-language sentiment analysis research.

2 Problem Statement

Existing sentiment analysis tools and pretrained NLP models are predominantly trained on formal, monolingual English text and do not generalize well to the code-mixed, informal, and often grammatically inconsistent language used by Nepali online shoppers. Reviews on platforms such as Daraz frequently mix English vocabulary with romanized Nepali words and phrases within the same sentence, use inconsistent spelling of Nepali words (since there is no standardized romanization), and contain short, noisy, and sometimes sarcastic text.

As a result, sellers and platform managers lack a reliable automated way to:

- Determine whether the overall sentiment of a review is positive, negative, or neutral when the text is code-mixed.
- Identify the dominant topics or product aspects (e.g., price, delivery, product quality, packaging, seller behavior) being discussed across large volumes of reviews.
- Aggregate this information into a form that can support data-driven decisions.

This project addresses the problem of building and evaluating an NLP pipeline capable of performing sentiment classification and topic extraction on this Nepal-specific, code-mixed customer review dataset.

3 Objectives

The primary objectives of this project are:

1. To collect and preprocess a dataset of customer reviews scraped from Daraz Nepal, with a pipeline that specifically handles code-mixed English and romanized Nepali text.

2. To build and compare sentiment classification models, classical machine learning baselines (TF-IDF with Logistic Regression/SVM) versus transformer-based multilingual models for classifying reviews as positive, negative, or neutral.

3. To apply topic modeling techniques (e.g., LDA and/or embedding-based clustering) to identify the dominant themes and product aspects discussed across the reviews.

4. To evaluate the resulting models using standard metrics (accuracy, precision, recall, F1-score, confusion matrices) and to visualize sentiment and topic trends in an interpretable format.

4 Review of Literature/Technology

4.1 Related Work

Sentiment analysis of e-commerce customer reviews has been studied extensively, with the Daraz platform itself appearing as a data source in recent work. Islam et al. (2023) [1] built a Bangla-English sentiment dataset scraped from Daraz and found that a classical Support Vector Machine baseline remained competitive against deep learning models such as LSTM and Bi-LSTM. In a Roman-Urdu setting, Zeb et al. [2] similarly scraped multilingual (Roman-Urdu, Roman-Urdu-English, and English) opinions from Daraz.pk, showing that code-mixed, romanized text is a recurring challenge across Daraz's different regional markets.

For Nepali-language NLP, Singh et al. (2020) [3] curated a targeted aspect-based sentiment analysis dataset for Nepali with mBERT and BiLSTM baselines, directly relevant to the feature-level analysis this project intends to perform. Pahari and Shimada (2023) [4] studied Nepali-English code-switching on social media and released a dedicated code-switched sentiment dataset, confirming that Nepali speakers frequently alternate between English and Nepali when expressing opinions online, a pattern this project expects in Daraz Nepal reviews as well.

On the topic-modeling side, Ray and Chakrabarti (2020) [5] combined LDA-based topic modeling with aspect-specific sentiment classification, reporting accuracy of 74-81% across aspects, and Karagoz et al. [6] extended this with Sentence-Segment LDA (SS-LDA) to address data sparsity on short review text. These studies motivate combining sentiment classification with topic modeling rather than treating the two as independent tasks.

4.2 Related Theory

Sentiment Analysis

Sentiment analysis is the task of determining the polarity (positive, negative, or neutral) expressed in a piece of text. Traditional approaches rely on lexicon-based scoring or classical machine learning classifiers trained on bag-of-words or TF-IDF features, while modern approaches use deep learning and transformer-based contextual embeddings to capture semantic and contextual nuance.

Topic Extraction

Topic extraction identifies and isolates salient themes embedded within a text corpus, distilling the latent semantic structure of large collections of documents. Latent Dirichlet Allocation (LDA) and Non-negative Matrix Factorization (NMF) are widely used probabilistic and matrix-factorization approaches, while more recent embedding-based approaches (e.g., BERTopic) combine transformer embeddings with clustering algorithms to produce more semantically coherent topics without requiring prior knowledge of the categories present.

LSTM

Long Short-Term Memory (LSTM) is a recurrent neural network architecture designed to mitigate the vanishing-gradient problem found in traditional RNNs. It captures long-term dependencies in sequential data through a memory cell and gating mechanisms, making it well suited to modeling the sequential structure of review text.

BERT

Bidirectional Encoder Representations from Transformers (BERT) is a pre-trained transformer-based language model that captures bidirectional context, allowing it to represent the meaning of a word based on both the words that precede and follow it. Multilingual variants such as mBERT and XLM-R extend this bidirectional, self-attention-based architecture across many languages simultaneously, which makes them a practical choice for the code-mixed English/romanized-Nepali text expected in this project.

4.3 Code-Mixed and Low-Resource Language Processing

Nepali is a low-resource language for NLP compared to English, and romanized Nepali text introduces additional challenges such as inconsistent spelling (there is no single standardized romanization scheme) and a lack of large annotated corpora. Prior work in Nepali NLP has explored transliteration, language identification for code-mixed text, and the use of multilingual transformer models as a practical way to handle low-resource and code-mixed languages without requiring large amounts of Nepali-specific labeled data.

4.4 Relevant Tools and Libraries

The project will draw on established open-source NLP tooling, including:

- **Scikit-learn** for TF-IDF vectorization and classical classifiers.
- **Gensim** for LDA topic modeling and word embeddings.
- **Hugging Face Transformers** for multilingual transformer models (e.g., XLM-R, mBERT).
- **Pandas / NumPy** for data handling and analysis.
- **Matplotlib / Seaborn / Plotly** for visualization.
- **BeautifulSoup / Selenium** for dataset collection from Daraz.

4.5 Gap Identified

While Daraz reviews have already been used as a data source in Bangla and Roman-Urdu sentiment studies, and Nepali sentiment/aspect-level datasets exist independently, there is comparatively little published work that combines the two: feature/aspect-level sentiment analysis together with topic modeling on Daraz *Nepal* reviews written in code-mixed English and romanized Nepali. This project aims to fill that specific gap by building and evaluating a pipeline adapted to, and evaluated on, Daraz Nepal review data.

5 Methodology / Approach

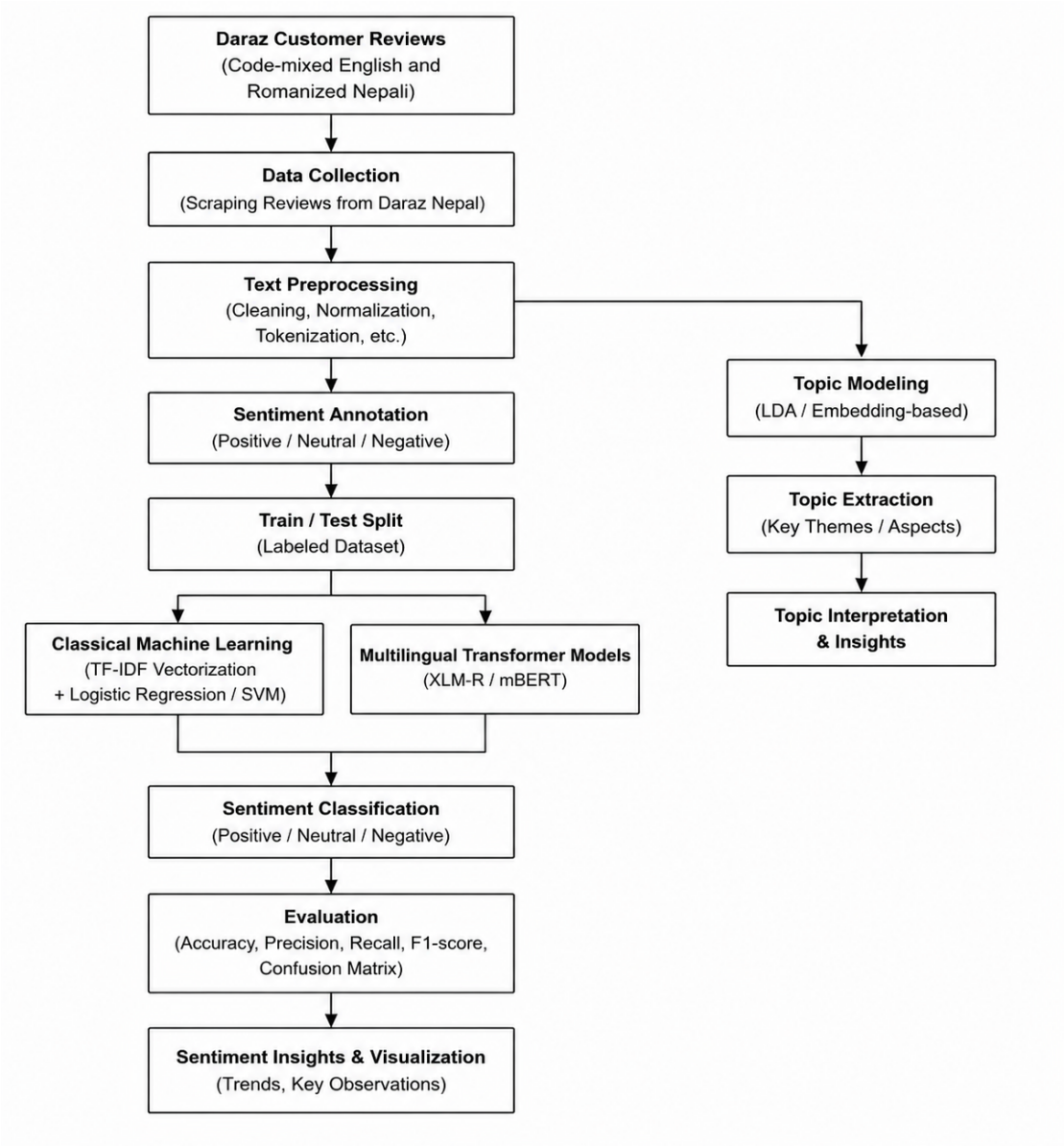


Figure 1: Overall workflow of the proposed sentiment and topic analysis system.

5.1 Data Collection

Customer review data will be scraped from the Daraz Nepal e-commerce platform across multiple product categories to ensure vocabulary and topic diversity. Collected fields will include review text, product category, and timestamp where available. A subset of the collected reviews will be annotated with sentiment labels (positive, negative, or neutral) for sentiment classification.

5.2 Data Preprocessing

- Cleaning of HTML artifacts, emojis, and irrelevant symbols.
- Normalization of romanized Nepali spelling variants where feasible.
- Language-mix handling: retaining code-mixed structure rather than forcing translation, since translation can lose sentiment-bearing nuance.
- Tokenization suited to multilingual/subword models.
- Removal of duplicate and low-information (e.g., single-word) reviews.

5.3 Exploratory Data Analysis(EDA)

Analysis of review length distribution, most frequent terms/n-grams, and the proportion of code-mixed versus monolingual reviews, to inform preprocessing and modeling decisions.

5.4 Sentiment Classification

- **Baseline models:** TF-IDF features with Logistic Regression and/or Support Vector Machine classifiers.
- **Advanced model:** Fine-tuning a multilingual transformer model (e.g., XLM-R or multilingual BERT) on the labeled review data for 3-class sentiment classification (positive/negative/neutral).
- Comparative evaluation between the classical baseline and the transformer-based model.

5.5 Topic Modeling

- Application of LDA for keyword-based topic discovery.
- Optional application of embedding-based clustering (e.g., BERTopic-style approach) as a comparative method.
- Manual labeling/interpretation of discovered topics (e.g., "delivery experience," "product quality," "pricing," "packaging").

5.6 Evaluation

Sentiment classification will be evaluated using accuracy, precision, recall, macro-F1, and confusion matrices. Topic modeling quality will be assessed qualitatively (topic coherence and interpretability) and, where feasible, using coherence scores.

5.7 Visualization and Reporting

Final results will be presented through visualizations such as sentiment distribution charts, word clouds per topic/sentiment class, and topic-sentiment cross-tabulations, compiled into a final report and/or a simple interactive dashboard.

6 Expected Outcomes

- A cleaned and documented dataset of Daraz Nepal customer reviews suitable for NLP research.
- A working sentiment classification pipeline capable of handling English and romanized Nepali code-mixed text.
- A topic modeling component that surfaces the dominant themes discussed in customer reviews.
- A comparative evaluation between classical machine learning baselines and transformer-based models on this dataset.
- Visual summaries (charts, word clouds, dashboards) that make review sentiment and topic trends interpretable to non-technical stakeholders.
- A final project report and presentation documenting methodology, results, limitations, and future work.

7 Role of Each Team Member

Member	**Responsibilities**
Bikash Ghimire	Data collection and web scraping from Daraz Nepal; data cleaning and preprocessing pipeline; exploratory data analysis (EDA).
Himanshu Chandra	Sentiment classification models – baseline (TF-IDF + classical ML) and transformer-based (multilingual BERT/XLM-R) fine-tuning and evaluation.
Sumin Magar	Topic modeling (LDA/clustering-based topic extraction), visualization/dashboard development, and compilation of the final report and presentation.

8 Timeline

The project is planned over a duration of 3 months, as outlined below:

Duration	**Activities**
Week 1–2	Literature review, finalizing project scope, tool/library setup, initial Daraz scraping script development.
Week 3–5	Full-scale data collection from Daraz Nepal, data cleaning, preprocessing pipeline for code-mixed text, sentiment annotation, and EDA.
Week 6–8	Baseline sentiment model (TF-IDF + classical ML) development and evaluation; initial topic modeling (LDA) experiments.
Week 9–10	Transformer-based sentiment model fine-tuning (multilingual BERT/XLM-R); comparative evaluation against baseline.
Week 11	Visualization/dashboard development, topic-sentiment integration, and error analysis.
Week 12	Final report writing, results compilation, and presentation preparation.

9 References

[1] Islam, M. R. et al., "Sentiment analysis in multilingual context: Comparative analysis of machine learning and hybrid deep learning models," *Heliyon*, 2023.

[2] Zeb, A. et al., "Multi-Lingual E-Commerce Daraz.PK Sentiment Analysis Detector Using Machine Learning Techniques," 2023.

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[4] Pahari, N. and Shimada, K., "Language Preference for Expression of Sentiment for Nepali-English Bilingual Speakers on Social Media," *Proceedings of the 6th Workshop on Computational Approaches to Linguistic Code-Switching (CALCS)*, ACL, 2023.

[5] Ray, P. and Chakrabarti, A., "Aspect-Oriented Sentiment Analysis: A Topic Modeling-Powered Approach," *Journal of Intelligent Systems*, 2020.

[6] Karagoz, P. et al., "A new topic modeling based approach for aspect extraction in aspect based sentiment analysis: SS-LDA," *Expert Systems with Applications*, 2020.